

Carnet: 00225424

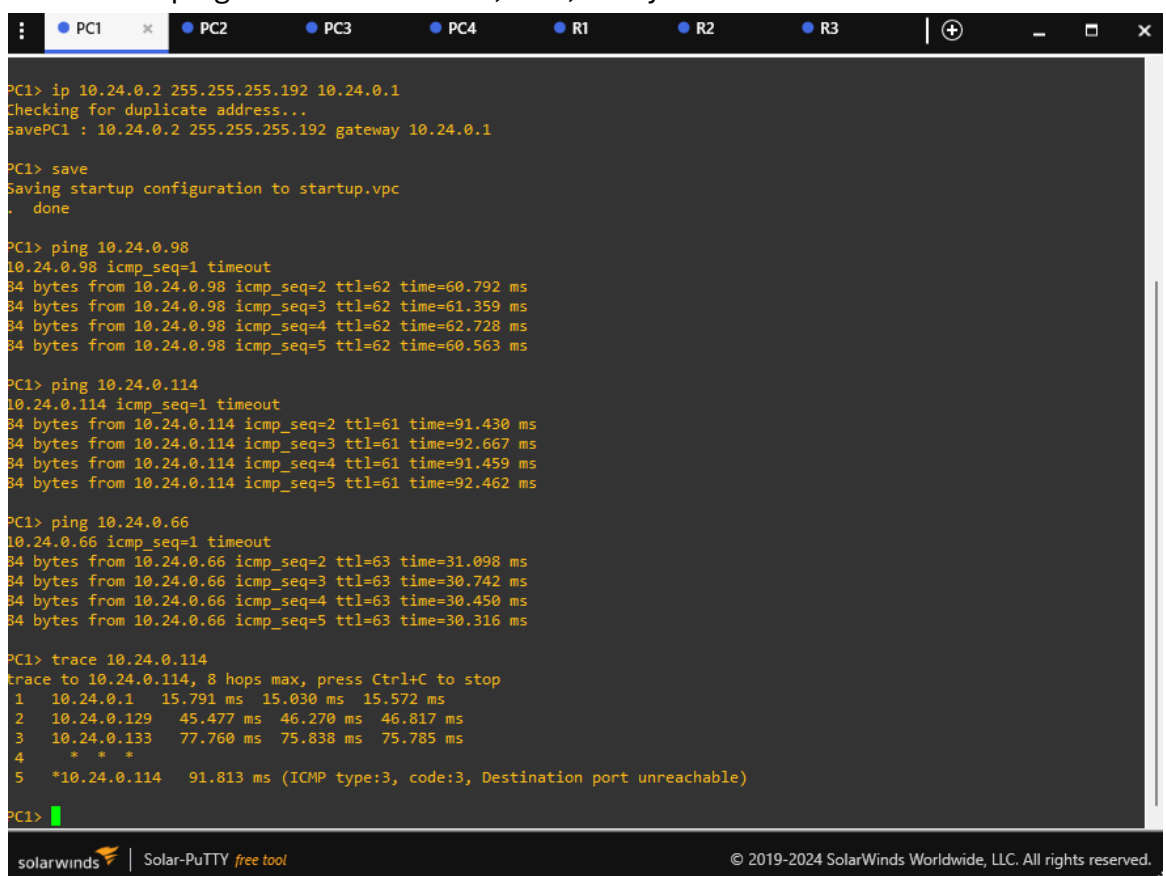
A: 00

B: 24

Red base asignada:

Ejercicio 1: VLSM y enrutamiento estático en GNS3 (2 puntos) Utilizando la red base personalizada 10.B.A.0/24, diseñe e implemente el direccionamiento IP de la topología indicada por el docente. Posteriormente, configure rutas estáticas para permitir comunicación entre todas las redes 10.24.0.

- Pruebas de ping exitosas entre PC1, PC2, PC3 y PC4.



```
PC1> ip 10.24.0.2 255.255.255.192 10.24.0.1
Checking for duplicate address...
savePC1 : 10.24.0.2 255.255.255.192 gateway 10.24.0.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> ping 10.24.0.98
10.24.0.98 icmp_seq=1 timeout
34 bytes from 10.24.0.98 icmp_seq=2 ttl=62 time=60.792 ms
34 bytes from 10.24.0.98 icmp_seq=3 ttl=62 time=61.359 ms
34 bytes from 10.24.0.98 icmp_seq=4 ttl=62 time=62.728 ms
34 bytes from 10.24.0.98 icmp_seq=5 ttl=62 time=60.563 ms

PC1> ping 10.24.0.114
10.24.0.114 icmp_seq=1 timeout
34 bytes from 10.24.0.114 icmp_seq=2 ttl=61 time=91.430 ms
34 bytes from 10.24.0.114 icmp_seq=3 ttl=61 time=92.667 ms
34 bytes from 10.24.0.114 icmp_seq=4 ttl=61 time=91.459 ms
34 bytes from 10.24.0.114 icmp_seq=5 ttl=61 time=92.462 ms

PC1> ping 10.24.0.66
10.24.0.66 icmp_seq=1 timeout
34 bytes from 10.24.0.66 icmp_seq=2 ttl=63 time=31.098 ms
34 bytes from 10.24.0.66 icmp_seq=3 ttl=63 time=30.742 ms
34 bytes from 10.24.0.66 icmp_seq=4 ttl=63 time=30.450 ms
34 bytes from 10.24.0.66 icmp_seq=5 ttl=63 time=30.316 ms

PC1> trace 10.24.0.114
Trace to 10.24.0.114, 8 hops max, press Ctrl+C to stop
 1  10.24.0.1  15.791 ms  15.030 ms  15.572 ms
 2  10.24.0.129 45.477 ms 46.270 ms 46.817 ms
 3  10.24.0.133 77.760 ms 75.838 ms 75.785 ms
 4  * * *
 5  *10.24.0.114 91.813 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>
```

- Prueba de conexiones entre routers

```

PC1 PC2 PC3 PC4 R1 R2 R3
*Mar 1 00:03:24.051: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:03:25.051: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config)#interface serial2/0
R1(config-if)#ip address 10.24.0.130 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:03:59.967: %LINK-3-UPDOWN: Interface Serial2/0, changed state to up
R1(config)#
*Mar 1 00:04:00.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R1(config)#
*Mar 1 00:04:22.291: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down
R1(config)#
*Mar 1 00:05:52.303: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
R1(config)#ip route 10.24.0.96 255.255.255.240 10.24.0.129
R1(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.129
R1(config)#ip route 10.24.0.132 255.255.255.252 10.24.0.129
R1(config)#exit
R1#
*Mar 1 00:22:32.975: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C       10.24.0.0/26 is directly connected, FastEthernet0/0
C       10.24.0.64/27 is directly connected, FastEthernet0/1
S       10.24.0.112/28 [1/0] via 10.24.0.129
S       10.24.0.96/28 [1/0] via 10.24.0.129
C       10.24.0.128/30 is directly connected, Serial2/0
S       10.24.0.132/30 [1/0] via 10.24.0.129
R1#
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```

```

PC1 PC2 PC3 PC4 R1 R2 R3
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface Serial2/1
R2(config-if)#ip address 10.24.0.134 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar 1 00:05:52.287: %LINK-3-UPDOWN: Interface Serial2/1, changed state to up
R2(config)#
*Mar 1 00:05:53.291: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/1, changed state to up
R2(config)#
*Mar 1 00:06:22.499: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/1, changed state to down
R2(config)#
*Mar 1 00:07:32.499: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/1, changed state to up
R2(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.130
R2(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.130
R2(config)#ip route 10.24.0.112 255.255.255.240 10.24.0.133
R2(config)#exit
R2#
*Mar 1 00:21:53.763: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.130
S       10.24.0.64/27 [1/0] via 10.24.0.130
S       10.24.0.112/28 [1/0] via 10.24.0.133
C       10.24.0.96/28 is directly connected, FastEthernet0/0
C       10.24.0.128/30 is directly connected, Serial2/0
C       10.24.0.132/30 is directly connected, Serial2/1
R2#
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```

```
PC1 PC2 PC3 PC4 R1 R2 R3
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#ip address 10.24.0.113 255.255.255.240
Mar 1 00:05:19.235: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
Mar 1 00:05:20.235: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#interface serial12/0
R3(config-if)#ip address 10.24.0.133 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
Mar 1 00:05:46.347: %LINK-3-UPDOWN: Interface Serial12/0, changed state to up
R3(config)#
Mar 1 00:05:47.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial12/0, changed state to up
R3(config)#ip route 10.24.0.2 255.255.255.192 10.24.0.134
Inconsistent address and mask
R3(config)#ip route 10.24.0.0 255.255.255.192 10.24.0.134
R3(config)#ip route 10.24.0.64 255.255.255.224 10.24.0.134
R3(config)#ip route 10.24.0.128 255.255.255.252 10.24.0.134
R3(config)#exit
R3#show
Mar 1 00:20:34.731: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
    10.24.0.0/26 [1/0] via 10.24.0.134
    10.24.0.64/27 [1/0] via 10.24.0.134
    10.24.0.112/28 is directly connected, FastEthernet0/0
    10.24.0.128/30 [1/0] via 10.24.0.134
    10.24.0.132/30 is directly connected, Serial12/0
R3#
```

Ejercicio 2: VLSM y enrutamiento dinámico OSPF en GNS3 (3 puntos) Utilizando la red base personalizada 192.B.A.0/24, diseñe e implemente el direccionamiento IP de la topología indicada por el docente. Posteriormente, configure enrutamiento dinámico mediante OSPF área 0 para permitir comunicación entre todas las redes.